

An Introduction to Typing Tutors

From Shorthand to Letter Invaders:
The Machines, Exercises, and Games That Taught Us to Type

Flyxion
Independent Researcher

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Preface: What a Tutor Teaches

A keyboard layout specifies which movements are possible. A typing tutor specifies which sequences of movements become thinkable, habitual, and eventually invisible. This book is about the second specification, which is easy to overlook because it is rarely stated. It lives in exercises.

The Method

Every typing tutor contains a theory of skill, whether or not its authors wrote one down. A lesson that adds one key at a time assumes that skill accumulates by small extensions of mastered territory. A lesson that rewards speed regardless of errors assumes that velocity is the primary quantity. A lesson built from digraphs treats language as a distribution of transitions, while one built from isolated letters treats the keyboard as a set of positions. These commitments can be recovered from the exercises themselves.

The method has three steps.

1. Treat the exercise as primary evidence and describe it exactly: what is on the page, in what order, repeated how many times.
2. State the generative rule that would reproduce it. Chapter 8 does this for one printed curriculum, whose lessons admit a growing key set K_i and therefore a growing available vocabulary

$$V_i = \{ w \in V : \Sigma(w) \subseteq K_i \}.$$

3. Compare rules across tutors, layouts, and media, so that a difference in output can be traced to a difference in the rule instead of being left as an impression.

What the Book Does Not Claim

Four limits are stated at the outset.

The paper tutors of Chapter 8 rest on a small number of photographed pages from two printed sources, and the pages do not settle how the sources were used together. Where the pages are silent, the text marks its statements as inference. Where a line is legible only as show-through from the reverse of a page, it is not quoted as text.

The history in Parts I to IV is organized around one question, namely how inscription became a repeatable sensorimotor procedure. It is not a complete history of writing machines, and it omits much that a history of any single device would require. Dates and attributions that could not be checked while drafting carry a **[Source needed]** flag.

Claims about layouts, Dvorak above all, have been contested for decades. The book reports the dispute without settling it. The curricula of Part VII compare layouts under identical pedagogy precisely so that the comparison does not depend on the disputed historical claims.

The DOS material of Part VI is specified as a framework and an evidence schedule. Statements about particular programs are to be filled from emulated sessions, manuals, and screenshots, and are flagged until then.

The Argument in Outline

Each part removes an assumption that seems obvious. Part I shows that writing need not preserve ordinary spelling. Part II shows that a fixed array of keys can stand in for the moving hand, and that institutions, not only machines, trained the body to use it. Part III shows that a keypress need not make a mark on paper, and that “typing” has named several distinct kinds of work. Part IV shows that a keystroke need not be a single key, and that the arrangement of keys is a variable. Part V supplies the vocabulary: implicit theories of skill, a taxonomy of tutors, and a learner state $S = (A, V, H, E, T, R)$ that explains why two typists with the same words per minute need different lessons. Part VI turns to the DOS era, when tutors became legible enough to read as machines. Part VII presents the exercises themselves, and Part VIII specifies a tutor that keeps a complete ledger of practice.

The last assumption to go is that an exercise is neutral because its content is trivial. A line such as **a sad lad asks dad**; is not merely poor prose. It is the visible shape of a restriction on what may be said, and different restrictions produce different temporary Englishes.

How to Read It

The chapters are cumulative through Part V and can be read more freely afterward. Readers with a particular interest in software may begin with Chapter 8, which carries the method in its most concrete form, and return to the earlier history afterward. Mathematical notation is gathered in Appendix D, and every formal statement is accompanied by a plain-language account. Transcriptions of the exercise pages used as evidence appear in Appendix C, and Appendix F lists every source page with its status, so that any claim about a lesson can be checked against the lesson.

Three evidence labels recur. Text that describes what is printed on a page is observation. Text marked *Inference* draws a conclusion from what a page omits. Text marked *Design* describes a proposal, such as a planned curriculum, and must not be read as a report of what any historical tutor did.

The central contention is simple and is tested throughout: what a person can type is decided by the keyboard, but what a person learns to type is decided by the tutor.

Part I

Fast Writing

Chapter 1

The Measured Hand

Before writing speed could be trained it had to be measured, and before it could be measured it needed a unit. The history of typing tutors begins with the history of that unit.

1.1 Copying as Labor

For most of the history of the codex, reproducing a text meant copying it by hand, and copying was paid work. Payment by the piece is one of the oldest ways of turning a bodily rate into an economic quantity. The university stationers of medieval Europe are a familiar example: exemplars were divided into sections (the *pecia* system) and rented out so that several scribes could copy one work at once. **[Source needed: secondary source on the pecia system and its dates]**

The point for this book is structural. Piece rates and rented exemplars already contain the two components of every later tutor: a model to be copied, and a rule for judging the copy. What they lack is an explicit measure of rate per unit time, since the scribe was paid for the product and not for the pace.

1.2 The Unit of Speed

A rate needs a countable unit. Natural words vary too much in length to serve, so later practice fixed a conventional word of five characters, spaces included. Under this convention gross speed is

$$\text{WPM}_{\text{gross}} = \frac{c/5}{t},$$

where c is the number of characters typed and t the time in minutes. The convention matters because it converts a linguistic quantity, the word, into a mechanical one, the keystroke. Every timed test in this book, including the five-minute writings of Chapter 8, rests on it. Appendix B states the scoring formulas and the variations found in the sources.

1.3 Speed as Capacity

Treating speed as a stable property of a person, something a test can reveal and practice can raise, is a modern idea. It requires standardized text, a clock, and an examiner. Each of these reappears in the chapters that follow: the standardized text as the timed passage, the clock as the metronome, the phonograph record and the cassette, and the examiner as the supervisor who initials a test sheet.

Chapter 2

Shorthand Traditions

Shorthand is the oldest sustained attempt to design inscription around the limits of the moving hand. Its central move is to separate the recording of language from ordinary spelling.

2.1 Ancient and Early Modern Systems

The Tironian notes, associated with Marcus Tullius Tiro, the secretary of Cicero, are the classical reference point for abbreviated writing. **[Source needed: dating and scope of the Tironian notes as a system versus a repertoire of signs]** In England, Timothy Bright published a system of “characterie” in 1588, and later seventeenth-century systems, among them Thomas Shelton’s, were in wide use; Shelton’s system is the one associated with the diary of Samuel Pepys. **[Source needed: verify Bright 1588 and the Shelton attribution]**

2.2 Phonetic Systems

The nineteenth century brought systems built on sound. Isaac Pitman’s phonography (1837) writes what is heard, with shading to distinguish voiced from unvoiced consonants. John Robert Gregg’s light-line system (1888) reduced the dependence on shading and made outlines more cursive. **[Source needed: confirm first-publication dates against a standard history]**

2.3 What Shorthand Teaches a Typing Tutor

Shorthand instruction anticipated three features of later tutors.

- **Graded units.** Learners meet a small set of signs first and build words from them, so the lesson order is itself a specification of a growing symbol set.
- **Dictation as pacing.** Speed is developed by reading aloud at a stated rate, which is the ancestor of the metronome, the phonograph record, and the cassette tape.

- **The gap between transcript and record.** A shorthand note is not the text; it must be transcribed. That gap between a fast record and a finished document recurs whenever a system separates capture from rendering.

Chapter 3

Stenotype and Machine Shorthand

Machine shorthand replaced the moving pencil with a keyboard on which several keys are pressed together. The machine records sound, not spelling.

3.1 Development

Machine shorthand is commonly traced to nineteenth-century American patents, with the twentieth-century Stenotype of Ward Stone Ireland (1911) as the design that established the pattern of a compact chorded keyboard printing on a paper strip. **[Source needed: patent dates and inventors; distinguish Bartholomew, Ireland, and later commercial machines]**

3.2 Institutional Setting

Court reporting is the paradigm application. The reporter's output is a paper strip in phonetic notation that must be transcribed into ordinary text, so the profession divides into capture and transcription, and its training divides accordingly into speed building and translation. **[Source needed: training standards and certification speeds, by period]**

3.3 Scope of This Chapter

This chapter is historical and institutional. The theoretical question, namely what happens to the notion of a keystroke when the primitive action is a chord, is reserved for Chapter 13, so that the two chapters divide the subject between them and do not repeat each other.

Chapter 4

Telegraphy

Telegraph operators were the first large occupational group whose skill was defined by a sustained, measured rate of sending and receiving.

4.1 Sending and Receiving

Samuel Morse's telegraph demonstrated a working line in 1844. [**Source needed: confirm date and route**] Operators were tested on both sending and receiving. Receiving initially involved reading a paper record, and skilled operators came to receive by ear.

4.2 A Labor Metric

Telegraph speed tests fixed the practice of stating skill as a number of words per minute, with a conventional word length so that different messages could be compared. The convention traveled with the operators. Many early typewriter operators had telegraph backgrounds, and the typewriter was used to transcribe received messages directly. [**Source needed: documentation of the telegrapher-to-typist pathway**]

4.3 Two Things Carried Forward

Two features of telegraph training recur in later tutors. The first is training by pacing signal: the learner keeps up with a sender. The second is a distinction between copying at speed and copying without error, a distinction that Chapter 15 will show is written into the printed instructions of at least one typing workbook.

Study Questions, Part I

1. Why does a rate of writing require a conventional unit of length? What does the five-character word gain and lose compared with counting natural words?
2. Shorthand and typing both separate capture from a finished document. Where does each place the cost of the gap?
3. Compare piece-rate payment for copying with a modern words-per-minute test. What is measured in each case, and who is the examiner?
4. Which features of telegraph training appear in the cassette-paced practice of Chapter 8?

Part II

The Typewriter

Chapter 5

The Fixed Character Array

The typewriter replaced continuous handwriting with selection from a fixed set. Once writing is selection, it can be taught as a table of positions.

5.1 From Writing Ball to Sholes

Several mechanical writing devices preceded the commercially successful typewriter. The machine developed in Milwaukee by Christopher Latham Sholes with Carlos Glidden and Samuel Soule was patented in 1868 and manufactured under E. Remington and Sons from 1873. **[Source needed: verify patent year and manufacturing start against a museum or patent source]** Type bars struck the paper from below, so the operator could not see what had just been typed, a condition that shaped early training: the learner had to trust position and not appearance.

5.2 The Consolidation of QWERTY

The QWERTY arrangement stabilized in the 1870s and became the default through manufacturer competition and the growth of trained typists. The popular explanation that the order was designed to prevent adjacent type bars from jamming is widely repeated and has been disputed in the literature. **[Source needed: cite the dispute; do not adjudicate here]** For the purposes of this book what matters is that a large population trained on one arrangement created a cost to changing it, a cost that Chapter 12 examines through the Dvorak case.

5.3 The Shift Key

The addition of a shift mechanism allowed one key to produce two characters. The consequence for pedagogy is that capitals are a separate skill: the learner must coordinate a second hand with the first. The binder examined in Chapter 8 introduces capitals by alternating case within a drill line (AsDsWs), which isolates exactly this coordination.

Chapter 6

Touch Typing and the Commercial School

Touch typing is the practice of typing without looking at the keys, using all fingers according to a fixed assignment. It is not a property of the typewriter. It is a training doctrine.

6.1 The Ten-Finger Method

Accounts of the origin of touch typing usually mention the memorization of the keyboard by court stenographers and the 1888 speed contest in Cincinnati between Frank McGurkin and Louis Traub, which was reported as a victory for the touch method. **[Source needed: primary or high-quality secondary account of the 1888 contest and of earlier claimants to the method]** Whatever the precise history, the doctrine that emerged prescribes a home position, an assignment of every key to one finger, and a habit of not looking.

6.2 Home Row as Doctrine

Home-row doctrine defines a resting position for the fingers and treats every other key as a reach from it. This is why the first lesson of a large family of tutors is the same eight keys, and why the reach drills of later lessons are written as excursions from and returns to home. Chapter 8 shows this structure in two independent printed sources.

6.3 The Commercial School

The vocational school gave the doctrine an institutional form: a fixed curriculum, a supervised room, a machine of known type, and an examiner. The apparatus of Chapter 8 is a late instance of this form. **[Source needed: history of commercial schools and business colleges in the period]**

Chapter 7

Office Labor and Ergonomic Doctrine

Typing became an occupation, and an occupation acquires rules for the body.

7.1 The Typing Pool and the Contest

Large offices concentrated typists in pools, and typing contests and trade demonstrations publicized speed as a marketable skill. **[Source needed: sources on typing pools, contests, and their gendered labor history]** Speed figures functioned as credentials, which is why so many tutors report their results in words per minute even when accuracy would predict employability better.

7.2 Posture and Rhythm

Instructional literature prescribed posture, wrist position, and sitting height, and it also prescribed rhythm. Typing to music and to recorded pacing signals was promoted as a way of regularizing keystroke intervals. **[Source needed: documentation of rhythm-typing records and their dates]** The idea of an external pacing signal is a through-line of this book: the phonograph record, the metronome, the cassette tape of Chapter 8, the metronome of Chapter 19, and the falling letters of Chapter 20 are successive answers to one problem, namely how to make a learner keep a rate that the learner does not yet control.

7.3 Discipline

The typing lesson is a discipline in the ordinary sense: a routine that shapes a body by repetition under observation. The paper tutor makes this explicit, because its supervisor is the observer and its typed page is the record.

Chapter 8

The Paper Tutor

This chapter reconstructs, from photographed pages, the rules of two printed typing curricula. Neither contains a processor. Each is nonetheless a program in the sense that matters here: an ordered sequence of instructions, a fixed data format, and a runtime made of people and machines.

8.1 Two Printed Sources

Two distinct sources are used. They must be kept apart.

The Academy pages. Pages bearing the imprint of Academy of Learning, among them pages 1, 11, 12, 20, 21, 31 and 34, with “Revised 07/92” printed on page 34. The pages are organized as numbered *Exercises*.

The workbook. A set titled *Keyboard Skill Building*, copyright Qwerty Learning Systems Inc. 1992, code KBD103/392, organized as twenty *Lessons* with a table of contents, instructions to students, progress tests, and cassette tape practice.

Both carry 1992 dates and both are typewriter-based, so they may have been used in the same institution. (*Inference: the shared date and medium suggest a common institutional setting; the pages do not say so*) The book does not treat them as one document. Whether they shared a physical binder, and how a learner moved between them, is unresolved and is recorded in Appendix F.

8.2 The Academy Pages

8.2.1 The apparatus

Page 1 is a labeled diagram of an electronic daisy-wheel typewriter, naming the platen knobs, paper bail, paper release lever, ribbon cassette, daisy wheel cassette, printing position indicator, and carrying handle. Its caption states that the learner’s typewriter should look the same or

very similar to the one illustrated, and that otherwise the learner should ask the supervisor for the manual for the machine in use.

Three consequences follow. Hardware is part of the specification. Variation in hardware is delegated to a human. And the pedagogy resides in the pages, while everything that adapts to circumstance resides in the supervisor.

8.2.2 The page specification

Each exercise page opens with a keyboard diagram in which the new keys are filled black and earlier keys remain outlined, a line naming the new keys and their fingers, a left margin of 1 to 1.5 inches, and single spacing. Exercise 1 assigns A, S, D, F to the left hand, semicolon, L, K, J to the right hand, and the space bar to the right thumb. The header fixes the physical output as well as the input: the learner is producing a correctly formatted page, which is the vocational deliverable, while acquiring the keystrokes.

8.2.3 The key order

The order in which keys are introduced is not alphabetical and not by letter frequency. On the pages available it is geometric.

Exercise	New keys	Position
1	A S D F ; L K J, space	home row, both hands
2	G H	home row, index stretch
6	W I	upper row, first reaches
13	Z	lower row, left little finger

Exercises 3 to 5 and 7 to 12 are not among the pages available. The home row is completed before any reach is attempted, and the rare letter Z arrives at Exercise 13, so its position follows finger geometry, not usefulness.

8.2.4 The early groups

Each exercise page is divided into numbered groups, each printed as three identical lines. On the pages for Exercises 1, 2, 6 and 13 the first six groups follow a common pattern, transcribed in Appendix C.

- G1.** A chain of the new keys, for example `asdf ;lkj` (Exercise 1) and `asdfg ;lkjh` (Exercise 2). The right hand is written in mirror order.
- G2.** The chain with one old key inserted, for example `asdfa ;lkj`; and `asdfgfa ;lkjhj`;. From Exercise 6 this group also carries the first shift practice: `ASDsWs ;LkIkL`. Exercise 13 gives `aSaZa ;L;P`;
- G3.** Either a scramble of the first group (`dafs k;jl`, Exercise 1) or short words (`wed kid was lie`, Exercise 6; `zip zoo zig zag`, Exercise 13).

- G4.** Word list. ass lad add alf; glad asks hash gala gald; ail law his owe; zinc: zulu zero jazz:.
- G5.** Longer words. flak alas fall asks; salad glass shall flash jaffa slash; were soil will hide; quiz doze size hazy:.
- G6.** A phrase or a list of longer words. a sad lad asks dad; closes Exercise 2, and heir wear like wore and zeal lazy fizz fuzz: close Exercises 6 and 13.

Group 3 of Exercise 1 contains one deliberate variant (k;jk among k;jl), so the line is not a perfect repetition.

8.2.5 The later groups

Two pages show that exercises continue past Group 6, and that the later groups are sentences.

Page 21 continues Exercise 6 with Groups 7 to 11. Group 7 is a word chain (wise lied wild wire lie), Group 8 is a short sentence (A wood fire is good; oil is worse for a fire; wood is for fire), and Groups 9 to 11 are longer sentences in which each content word draws on the new keys:

Group 9	We killed wild doe for a while for food; Will would ride well
Group 10	A fellow said he walked a high hill; he grew riled if we rode;
Group 11	Ilse who is a redhead is a selfish girl who likes a wild life;

Page 31 belongs to an exercise between Exercises 6 and 13 whose number is not established from the photograph. Its Groups 7 to 13 run from word chains to sentences:

Group 9	Peter and Paul ran quickly to the printers for a paper:
Group 10	King Pepe snapped his fingers as the Pekinese pup passed by;
Group 12	I will quit the room again if he does not answer all queries:
Group 13	The Queen was presented with two quills at the banquet today;

So the group count varies by exercise, and the six-group ladder is the early portion of a longer sequence that ends in continuous sentences. The last group of each observed exercise is a sentence or a run of sentences, and the sentences are selected to exercise the new keys. The page for Exercise 13 and other pages show faint lines through the paper from the reverse side, and these are not quoted as text.

8.3 The Sandwich Drill

Both sources teach a reach by the same device: the reached key is placed between two presses of the home-row key that normally covers it. In the Academy pages the device appears as asdsws ;lkikl (Exercise 6, Group 1). In the workbook it opens the second block of Lesson 1:

```
aqa sws ded frf ftf fgf ;p; lol kik juj jyj jhj aza sxs
dcd fvf fbf ;/; 1.1 k,k jmj jnj
```

and is immediately followed by a line pairing each drill with a word that contains it (**aqa** **aqua**, **sws** **sail**, **ded** **dead**, **frf** **rain**, **ftf** **tall**, **fgf** **gulf**). The sandwich drill isolates the transition from home to reach and back, which is the unit that the learner state of Chapter 17 calls transition competence.

8.4 A Formal Model of the Academy Pages

Let K_i be the set of keys admitted by exercise i , with

$$K_0 \subset K_1 \subset \dots \subset K_n,$$

and let $N_i = K_i \setminus K_{i-1}$ be the new keys. For a word w let $\Sigma(w)$ be the set of characters it requires. The vocabulary available to the author of exercise i is

$$V_i = \{ w \in V : \Sigma(w) \subseteq K_i \}.$$

Two regimes appear. In the earliest exercises the binding constraint is admissibility itself: V_1 is so small that the lesson is nearly all of it, which is why **alf** and **gald** appear. In later exercises V_i is large and the author selects for salience,

$$V_i^{\text{new}} = \{ w \in V_i : \Sigma(w) \cap N_i \neq \emptyset \},$$

which is why Exercise 13 is a catalogue of words containing Z and why every word in Exercise 6 contains W or I. The constraint migrates from what can be spelled to what will exercise the new key.

Each group G_j has a cyclic unit u_j and is printed as three lines, each a window of width ω on the endless repetition of the unit:

$$\text{line}_j = \text{prefix}_\omega(u_j^\infty), \quad G_j = (\text{line}_j)^3.$$

Lines end mid-unit (**asd**, **daf**, **lag**), so the carriage return never coincides with a unit boundary and the typist has no natural place to pause. An exercise is the sequence $E_i = (G_1, \dots, G_m)$ with m varying by exercise. What is not on the pages is the advancement rule, and the judgment J that moves the learner from E_i to E_{i+1} is not written in the Academy pages available.

8.5 The Workbook

8.5.1 Lesson template

The workbook uses a different template. Its lessons do not restrict the alphabet: Lesson 1 opens with a speed drill of ordinary full-alphabet sentences. Each lesson is a fixed sequence of sections with a stated number of repetitions.

Section	Repeats	Content in Lesson 1
Speed Drill	3	four ordinary sentences, each typed three times
Keyboard Drill	2	sandwich drills, then drills paired with words
Alphabet Drill	2	alphabet in runs, then one three-letter word per letter, then a full-alphabet sentence
Word families, phrases, sentence	1	inflected word families, slash-separated phrases, four sentences
Tape practice	once	paced typing at a stated rate
Timed writings	2	five-minute tests

The repetition counts fall (3, 2, 2, 1) as the material becomes longer and more varied, a schedule different from the uniform three lines of the Academy pages.

8.5.2 The lesson schedule

The table of contents gives the sequence of the twenty lessons. Every fifth lesson is a test lesson with no tape practice or timed writings; the others end with tape practice at a rate that steps upward in blocks.

Lessons	Drill emphasis	Tape rate
1 to 4	keyboard; word families, prefixes, suffixes	25 wpm
5	practice for and administration of Progress Test 1	none
6 to 9	number drill; suffixes, double letters	30 wpm
10	Progress Test 2	none
11 to 14	symbol drill; word families, prefixes, suffixes, double letters	35 wpm
15	Progress Test 3	none
16 to 19	alphabet sentences; alternate-hand, one-hand words, spelling	40 wpm
20	Final Progress Test	none

The workbook therefore has a numeric speed curriculum: a stepped schedule of five words per minute per block, delivered by a cassette tape acting as a pacer. This is absent from the Academy pages.

8.5.3 Two policies on errors

The workbook gives its errors policy in printed instructions. The general instructions say that typing errors are *not* to be corrected. Drill lines are then marked and counted: the sample page shows errors circled by the student and a count written beside each set of lines, and a stated scale reads zero to three errors as excellent, four to seven as satisfactory, and eight or more as a signal to practice further. The phrase lines contain diagonal slash marks which the learner must not type; the instructions say they are present to make the learner think and type in groups of words.

The cassette practice reverses the priority. Its instructions tell the learner not to worry about the number of errors and to concentrate on keeping up with the tape, so that speed is developed. The one workbook therefore contains two theories at once: accuracy is the object of the drill lines and pace is the object of the tape. Chapter 15 develops this.

Because errors are not corrected, the page records every uncorrected error by policy, not only by the limits of the machine. This restores, for the workbook, the claim that a typed page is a complete record of uncorrected error. It does not record timing, and it cannot record corrections, because none are permitted.

8.5.4 Timed writings, progress tests, and the completion sheet

Timed writings are five minutes long. The learner practices each passage once, then is timed for five minutes, repeated twice. Numbers at the right of the passage give the cumulative word count for each line, and a second set of numbers beneath the timing is used when the test ends mid-line; neither set is to be typed. Paragraphs are indented five spaces and line endings must match the passage so that the test can be scored. The learner uses a clean sheet, types a name on the sixth line from the top, triple spaces, sets double spacing, and types one timing on the front of the sheet and the second on the back.

The printed text on the back of these instruction pages shows through the paper and is only partly legible. It appears to say that the learner may do each test twice, hands in the better one to be checked and initialed, and, when it is returned, enters the speed on the Lesson Completion Sheet and files the test as the last sheet of that set of lessons. (*Inference: this reading is from show-through and must be checked against the reverse page before quotation*)

8.6 The Judgment *J*

In the Academy pages the advancement rule *J* is unwritten, and its absence is evidence that it lives in a person. In the workbook *J* is partly documented. A learner completes lessons in sequence; scores drill lines against a stated scale; produces two five-minute timings per lesson; has the better one checked and initialed by a supervisor; enters a speed on a completion sheet; and is tested at Lessons 5, 10, 15 and 20. What remains unknown is whether a threshold on the completion sheet gated advancement. The Lesson Completion Sheet and a Progress Test page are needed to settle this. **[Source needed: Lesson Completion Sheet and one Progress Test page]**

8.7 The Paper Tutor as a Distributed System

Assembled, the workbook and the Academy pages together have components that map onto the parts of a program.

Program component	Paper tutor
Instruction sequence	ordered exercises or lessons
Data format	header block, groups or sections, repeated lines
Execution engine	learner's hands and the typewriter
Pacing signal	cassette tape (workbook only)
Output record	typed page, uncorrected
Error counter	the learner, marking and counting
Conditional branch	supervisor's check, initials, and test results
Exception handler	supervisor, for machine variation
Persistent state	the filed stack of lessons and the completion sheet

The program is executed across several substrates at once. Its state after any session is the stack of typed pages, which records every uncorrected error but no intervals between keystrokes.

8.8 What Passed to the DOS Tutor

Automated tutors preserved the graded key order, the per-lesson lexicon, the highlighted keyboard diagram, and short-unit repetition, and they absorbed the pacing tape into the computer. They added measured speed, immediate error feedback, and stored scores. They generally discarded the page. A DOS tutor commonly keeps a number where the paper tutor kept an artifact, and the practice ledger of Chapter 27 is a deliberate return to the page.

Figures

- Figure 8.1: typewriter diagram and supervisor caption (Academy p. 1).
- Figure 8.2: Exercise 1, home row (p. 11).
- Figure 8.3: Exercise 2, G and H (p. 12).
- Figure 8.4: Exercise 6, W and I, with first shift alternation (pp. 20, 21).
- Figure 8.5: Exercise 13, Z, with the “Revised 07/92” imprint (p. 34).
- Figure 8.6: a later exercise ending in sentences (p. 31).
- Figure 8.7: workbook instruction pages (drill lines, cassette practice, timed writings).
- Figure 8.8: workbook table of contents, Lessons 1 to 20.
- Figure 8.9: workbook Lesson 1, page 1-1.

Study Questions, Part II

1. Explain why a machine on which the typist cannot see the line just typed favors training by position over training by appearance.
2. State the sandwich drill as a rule. Why does it isolate the transition between home and reach?
3. Using the transcriptions in Appendix C, compute V_i^{new} for Exercise 6 and state which words on page 21 are outside it.
4. The workbook forbids correcting errors. List two things this policy makes observable and two that it hides.
5. Draw the supervisor's decision procedure for the workbook as far as the pages document it, and mark each edge whose rule is not printed.

Part III

Composition Keyboards

Chapter 9

The Linotype

The Linotype shows that “typing” has never named a single kind of work. An operator at the keyboard does not produce a page. The keyboard releases brass matrices that assemble into a line of type, which is then cast as one piece.

9.1 Mechanism

Ottmar Mergenthaler’s Linotype was developed in the 1880s and entered newspaper production in the middle of that decade. **[Source needed: dates and first installations]** Pressing a key releases a matrix from a magazine; matrices and spacebands accumulate to fill a line; the line is cast in metal and the matrices are returned to their channels for reuse.

9.2 A Keyboard Not Built on QWERTY

The Linotype keyboard is not arranged like a typewriter’s. Its lowercase keys follow an order derived from letter frequency, the source of the familiar nonsense phrase “etaoin shrdlu” that appears in printing lore when an operator runs a finger down the first columns to fill a line. **[Source needed: verify the arrangement and the origin of the phrase]** A training curriculum for this keyboard could not reuse the typewriter’s home-row lessons, and the operator’s speed was governed as much by the machine’s cycle as by the hand.

9.3 Consequences for Tutoring

Two points matter for later chapters. First, the layout is evidence that the arrangement of keys is a design variable chosen for a purpose. Second, the operator’s error is different in kind: a wrong key produces a wrong matrix that must be discarded or fixed in the line, so errors have material cost in metal and time.

Chapter 10

Monotype and Separated Keying

The Monotype system, developed by Tolbert Lanston, divides composition into two machines. **[Source needed: dates]** A keyboard perforates a paper ribbon, and a separate caster reads the ribbon and casts individual characters.

10.1 The Ribbon as Intermediate Record

Separating keying from casting inserts a record between the hand and the metal. The ribbon can be inspected, corrected, and re-run. It also carries the information needed to justify a line, so the operator's task includes tracking the space remaining in the line.

10.2 Why It Matters Here

The ribbon is an early instance of a persistent record of keystrokes that exists apart from the finished product. The paper page of Chapter 8 and the practice ledger of Chapter 27 belong to the same family, and Monotype shows that the family predates computing.

Chapter 11

Keypunches and Teletypes

The keyboards of data processing detach the visible mark from the mechanism.

11.1 Keypunches

A keypunch translates each keystroke into a pattern of holes in a card. The operator's product is a card that no reader can interpret without a machine, though many keypunches also printed the character along the top. Eighty-column cards became the dominant format in data processing. **[Source needed: dates and vendors]**

11.2 Teleprinters

Teleprinters and terminals transmit codes, and the mark is produced elsewhere. Later character encodings standardized what each keypress means. **[Source needed: ASCII dates and the relationship between teleprinter codes and computer keyboards]**

11.3 Emitting a Code

With the computer keyboard the typist no longer strikes a letter onto paper but emits a code that is interpreted somewhere else. That detachment is what makes a typing tutor possible in the sense of this book: the same computer that receives the trained gesture can observe, score, and reshape it.

Study Questions, Part III

1. What kinds of error are costly on a composition keyboard but not on a typewriter?
2. A perforated ribbon is a persistent record of keystrokes. Name one respect in which it is like the typed page of the paper tutor and one in which it differs.
3. Why does the detachment of mark from mechanism make an automatic tutor possible?

Part IV

Alternative Layouts

Chapter 12

Dvorak

The Dvorak Simplified Keyboard was patented in the 1930s by August Dvorak and William Dealey. **[Source needed: patent year and details]** It is the best-known alternative to QWERTY and the layout against which the curricula of Part VII are compared.

12.1 The Layout

In the standard arrangement the rows are as follows, with punctuation shown as it appears.

Row	QWERTY	Dvorak
Top	q w e r t y u i o p	' , . p y f g c r l
Home	a s d f g h j k l ;	a o e u i d h t n s
Bottom	z x c v b n m , . /	; q j k x b m w v z

Dvorak places all five vowels and the frequent consonants D, H, T, N, S on the home row, and the design principles usually cited are alternation between hands and concentration of work on the strongest fingers. Only two letters, A and M, occupy the same position in both layouts.

12.2 The Disputed Record

Claims that Dvorak is markedly faster have been repeated for decades and contested with equal persistence, and the evidence includes mid-century studies and later reappraisals whose conclusions differ. **[Source needed: cite the principal studies and reappraisals without endorsing either side]** This book does not adjudicate. The comparison of Part VII is designed so that it does not depend on the outcome.

12.3 Position Matching

Because both layouts are laid out on the same physical keys, any QWERTY lesson can be transposed to Dvorak by position: the key in each physical location is replaced by the key that

occupies that location in Dvorak. The letters of the mapping are these.

Row	QWERTY key	Dvorak key at that position
Top	Q W E R T Y U I O P	' , . P Y F G C R L
Home	A S D F G H J K L ;	A O E U I D H T N S
Bottom	Z X C V B N M , . /	; Q J K X B M W V Z

Applied to the Academy Exercises 1, 2, 6 and 13 of Chapter 8, the mapping gives the following.

Exercise	QWERTY new keys	Dvorak by position	Consequence
1	A S D F ; L K J	A O E U S N T H	vowels and four consonants; no semicolon
2	G H	I D	completes the home row
6	W I	comma, C	one new key is punctuation
13	Z	semicolon	a lesson with no letter at all

This is the cost of position matching, and it is the subject of Chapter 24: the same geometry yields a different lexicon at each stage, and at some stages no lexicon at all.

12.4 Matching Letters Instead

The alternative is to match content, teaching the same letters in the same order and letting the geometry differ. That preserves the vocabulary V_i at each stage but breaks the mapping between lesson number and keyboard region. Chapters 24 and 25 present both and compare them.

Chapter 13

Chorded Input and the Unit of Action

Chapter 3 described where machine shorthand came from. This chapter asks a different question: what happens to a tutor when the primitive action is no longer a single key?

13.1 Chords and Strokes

In a chorded system a character or syllable is produced by pressing several keys at once. The unit of action is the *stroke*, a set of keys pressed and released together. Let k be the number of keys and $s \subseteq \{1, \dots, k\}$ a stroke. The number of distinct strokes available on a board of k keys is $2^k - 1$, which is why a small chorded keyboard can address a large vocabulary.

13.2 What Changes for the Curriculum

In a single-key tutor the lesson order is an order over keys. In a chorded tutor it is an order over strokes, and strokes have structure: they overlap, so learning one stroke changes the cost of learning another. A curriculum can exploit the structure, for example by introducing strokes that share fingers together.

The error model changes as well. A wrong key in a single-key system is one wrong character. A wrong chord may differ from the intended one by a single finger, so the natural error metric is a distance between sets, for instance

$$d(s, s') = |s \triangle s'|,$$

the number of keys by which two strokes differ. The learner state of Chapter 17 has to be redefined with strokes in place of keys, and the transition term becomes a matrix over pairs of strokes.

13.3 Phonetic Briefs

Court-reporting theories also compress high-frequency words and phrases into single strokes, called briefs. A tutor for such a system teaches a dictionary as well as a motor skill. **[Source needed: a specific stenotype theory and its learning sequence]**

13.4 The Pedagogical Primitive

The finding of this chapter is a point of method. Single-key tutors treat the key as the unit of both action and error. The unit of action can differ from the unit of the tutor's display, so the definition of the tutoring primitive is a design decision that every tutor makes, whether or not it says so.

Chapter 14

One-Handed and Ergonomic Layouts

The relation between hand and keyboard is not fixed, and a substantial body of designs reshapes it.

14.1 Split and Staggered Boards

The row stagger of a standard keyboard descends from the mechanics of the typewriter's linkages, not from the geometry of the hand. Split keyboards separate the halves, and columnar layouts align keys with the movement of the fingers. **[Source needed: representative designs and dates]**

14.2 One-Handed Arrangements

One-handed layouts serve users with a single functioning hand and can also be used to reduce load. Some arrangements fold a standard layout onto one half, so that the other hand's keys are reached by a modifier. **[Source needed: sources on one-handed layouts, including designs derived from QWERTY]**

14.3 Implications for Tutors

Each layout implies a curriculum, and the curriculum must be regenerated when the geometry changes. This is the argument for treating the tutor's ladder as a template with three parameters: the key-introduction order, the lexicon, and the layout header, so that a new layout is a new instantiation and not a new tutor.

Study Questions, Part IV

1. Transpose Exercise 2 of the Academy pages to Dvorak by position, and state the new keys.
2. Why does position matching produce an empty V_i^{new} at some exercises? Give the two exercises in the design order.
3. On a chorded board with $k = 6$ keys, how many strokes exist? What is $d(s, s')$ for strokes $\{1, 2, 4\}$ and $\{1, 3, 4\}$?
4. Why can a tutor for a chorded system not treat the key as its unit of error?

Part V

A Theory of the Tutor

Chapter 15

Implicit Theories of Skill

Every tutor embodies a view of what skill is. This chapter lists the commitments that recur and shows how to read them from exercises.

15.1 Five Questions

A tutor answers each of the following, usually without saying so.

1. **What is the unit of practice?** An isolated key, a transition, a word, a phrase, or continuous prose.
2. **What comes first, speed or accuracy?**
3. **What is an error?** Something to be prevented, corrected, counted, or ignored.
4. **Where should attention be?** On the keys, on the hands, on the printed model, or on the meaning.
5. **What signal governs pace?** The learner, a supervisor, a metronome, a tape, or a machine.

15.2 Reading the Two Paper Sources

The sources of Chapter 8 answer these questions in different ways, and the workbook answers some questions twice.

Question	Academy pages	Workbook
Unit of practice	new key in a chain, then words spellable from the key set, then sentences	sandwich transitions, word families, phrases, sentences
Speed or accuracy	no speed stated on the pages available	accuracy scored on drill lines; pace prioritized on tape; speed recorded in timed writings
Error	implicitly a mismatch with a repeated model	marked, counted, never corrected
Attention	on the printed model and the keys	on the printed lines while listening to the tape; slashes chunk phrases
Pace signal	the learner and a supervisor	a cassette at a stepped rate

15.3 The Workbook's Double Theory

The workbook instructs the learner both to keep errors low on drill lines and, on tape practice, not to worry about errors while keeping up with the tape. These are not contradictory once seen as two phases of one theory: accuracy is established in a slow, self-paced phase and speed is then developed under an external pace at which some errors are tolerated. The tutor thereby encodes a sequential model in which accuracy precedes speed and speed is then imposed. (*Inference: the printed instructions state each policy separately; the sequential reading is an interpretation*)

15.4 Chunking by Typography

The slash marks in the phrase lines are the first explicit instruction about an attention unit. The learner is told not to type the slashes and that they are there to prompt thinking and typing in groups of words. This is a chunking device that moves the unit of practice from the word to the phrase without changing the text typed.

15.5 The Home Position as Theory

Both paper sources begin with home-row keys and treat later keys as excursions. The theory is that a stable resting position is the frame of reference for all movement. A tutor that instead began with the most frequent letters regardless of position would hold a different theory, in which language distribution, not hand geometry, sets the order.

Chapter 16

A Taxonomy of Tutors

The taxonomy classifies tutors by what they take learning to consist of. The categories overlap, and a tutor may occupy several.

16.1 Definitions

Copying tutor. Presents a model string and measures conformity.

Generative tutor. Constructs exercises from a restricted set of known keys, so the exercise is a function of the learner's position in the curriculum.

Diagnostic tutor. Infers weaknesses from errors, hesitations, or finger transitions.

Rhythmic tutor. Regulates cadence, by metronome, music, or a paced recording.

Transcription tutor. Introduces reading, memory, punctuation, formatting, or dictation.

Gamified tutor. Embeds the same gestures in a simulated emergency or contest.

Vocational tutor. Reproduces forms, correspondence, numerical entry, or typesetting work.

Corrective tutor. Concentrates practice on the learner's empirical error distribution.

16.2 Classifying the Paper Sources

Source	Classification
Academy pages	copying, with a generative layer (vocabulary constrained by K_i) and a vocational layer (page format and margins)
Workbook, drill sections	copying, with a corrective element, since the learner counts errors and is told to practice further above a threshold
Workbook, tape practice	rhythmic, by an external pacer at a stated rate
Workbook, timed writings	transcription and vocational, since scoring depends on line endings and paragraph indentation

The workbook's corrective element is coarse. The learner is directed to extra practice when errors are numerous, but the page does not tell the learner which keys or transitions were at fault. A diagnostic tutor would; the paper tutor leaves that to the supervisor.

16.3 A Test Case

Typing Tutor IV, discussed in Chapter 18, is described in catalog and retail records as testing a learner, identifying weak points, and then concentrating lessons on them. That is the definition of a corrective tutor, and it would place a commercial DOS-era program in the same class as the ledger-driven tutor proposed in Chapter 26. Whether the program in fact does so is an empirical question about its lesson-selection rule, answerable only from the running software.

16.4 A Category Not on the Original List

The workbook's alphabet sentences, in the later lessons, are sentences designed to contain every letter. They are constrained not by a restricted alphabet but by a completeness requirement, and they form a category of their own: the *coverage exercise*. The full-alphabet sentence in Lesson 1 (Only a few quite exceptional people do realize that there is very little value in just talking without being able to say something worthwhile.) contains all twenty-six letters. Coverage exercises are the dual of the restricted-alphabet exercise: one bounds the alphabet from above, the other requires it from below.

Chapter 17

Learner State and Lesson Selection

17.1 Observables

For a prompted character sequence $x_1, \dots, x_n$, a tutor may observe response times t_i , correctness indicators c_i , correction actions r_i , and, if the layout is known, the finger f_i assigned to each key. What a tutor can observe depends on the medium. The paper tutor observes only the final page, so it sees the uncorrected errors that remain and nothing about timing or corrections. A software tutor can observe all of the above.

17.2 Speed and Error Rates

Gross speed under the five-character convention is

$$\text{WPM}_{\text{gross}} = \frac{c/5}{t},$$

with c characters typed in t minutes. If e is the number of uncorrected errors, a common net measure is

$$\text{WPM}_{\text{net}} = \text{WPM}_{\text{gross}} - \frac{e}{t}.$$

Two error rates must be distinguished. The *uncorrected* rate is the fraction of final characters that are wrong. The *corrected* rate is the fraction of keystrokes that were errors, whether or not later repaired. A tutor that forbids correction, as the workbook does, measures only the first. A tutor that allows backspacing can measure both, and the difference between them describes the learner's correction behavior.

17.3 A Worked Example

Suppose a learner types 1500 characters in a five-minute timing and the final page contains 10 wrong characters. Then $c/5 = 300$ words, so $\text{WPM}_{\text{gross}} = 300/5 = 60$. The uncorrected errors number $e = 10$, so $\text{WPM}_{\text{net}} = 60 - 10/5 = 58$, and the uncorrected error rate is $10/1500 \approx 0.67\%$. If the same learner also made and repaired 40 further errors, a ledger would

show 50 erroneous keystrokes out of 1 540 character keystrokes (backspaces excluded), a rate of about 3.2% that the page alone could never reveal. The paper tutor of Chapter 8, which forbids correction, would report only the first figure.

17.4 Learner State

Crude words per minute collapses these observables into one number. A richer learner state is the tuple

$$S = (A, V, H, E, T, R),$$

where A is accuracy, V is velocity, H is hesitation structure, E is the error distribution, T is transition competence, and R is correction behavior. Operational definitions follow.

- A : uncorrected and corrected error rates.
- V : gross and net words per minute.
- H : the distribution of inter-key intervals, for example its spread and the frequency of intervals much longer than the median.
- E : the confusion matrix, with rows for intended keys and columns for struck keys.
- T : a matrix T_{ab} of mean latency for the transition from key a to key b , normalized by the latency expected from hand and finger alone.
- R : the rate of corrections and the delay before correcting.

17.5 Why Equal Speeds Are Not Equal Typists

Three typists with the same net speed can be pedagogically distinct.

Typist	Hesitation H	Correction R	Suggested emphasis
A	even intervals, slow	few corrections	pace training
B	bursts separated by long searches	few corrections	transition drills on the searched keys
C	fast, even	frequent backspacing	accuracy work at reduced pace

A sophisticated tutor ought to assign them different exercises.

17.6 Lesson Selection

Lesson selection can be stated as an optimization. If L is a candidate lesson and S the current state, the tutor selects

$$L^* = \arg \max_L \left[\mathbb{E}(\Delta S \mid L, S) - \lambda \text{Fatigue}(L) - \mu \text{Frustration}(L, S) + \nu \text{Transfer}(L) \right],$$

where $\lambda, \mu, \nu \geq 0$ weight fatigue, frustration, and transfer. The expected-improvement term is the substantive one; the others are regularizers. The formula is a specification, not a claim that any historical tutor computed it.

Definition 17.1. A tutor is *V-adaptive* if its lesson choice depends on the learner state only through the velocity component V . It is *fully adaptive* if the choice depends on other components of S .

Under this definition a game that raises falling speed after each success is *V-adaptive*, while a tutor that assigns transition drills for the keys on which latency is high is not. “Adaptive” ceases to mean merely “gets faster when you succeed”. A fixed-lesson paper tutor is not adaptive at all; whatever adaptation occurs happens in the supervisor.

Study Questions, Part V

1. For each of the five questions of Chapter 15, answer for a tutor you have used.
2. Give an example of two typists with the same net speed but different states S , and say which component differs.
3. Show that a tutor whose lesson choice depends only on words per minute is V -adaptive but not fully adaptive.
4. Why can a no-correction policy measure only the uncorrected error rate?

Part VI

The DOS Era

Chapter 18

Legibility

The tutors of the DOS period are treated here as a class, because their interfaces made the instructional machinery visible. Statements about individual programs are to be filled from emulated sessions, manuals, and screenshots and are flagged until then. **[Source needed: a list of programs to be documented, with edition, publisher, and source]**

18.1 What Legibility Means

A tutor is *legible* if a learner or observer can recover its pedagogical rule from the interface. The features to look for are these.

- A visible lesson menu that reveals the curriculum order.
- Explicit speed or accuracy thresholds for advancement.
- A keyboard diagram highlighting the finger and key.
- Audible or visible error signals.
- A local score table.
- Games whose rules can be understood at a glance.

18.2 Legibility and Graphics

Limited graphics forced the instructional model into view. When a tutor can draw only blocks and characters, the mapping from performance to consequence must be simple, and the rule is visible in the display. Later interfaces may look smoother while concealing whether adaptation is meaningful or merely cosmetic. This is a hypothesis to be tested against the documented programs, not a finding.

18.3 Documented Screens

Five screenshots are available. Four are now attributed, on the strength of their own title screens and the owner’s account of where each was obtained; one (S4) is not yet identified, and the book does not attribute it.

	Screen	Status
S1	<i>Typing Tutor III with Letter Invaders</i> , menu with two options, copyright Kriya Systems, Inc., 1984, published by Simon & Schuster, black and white	identified by its own title screen; obtained as Typing Tutor III (1984), per the owner; the catalog of Subsection 18.3.3 lists that title for IBM PC compatibles
S2	<i>Typing Tutor IV with Letter Invaders</i> , title screen, copyright 1983 and 1987, Kriya Systems, Sterling, Virginia, published by Simon & Schuster Software, “Press any key to begin”	identified by its own title screen; obtained as v1.0 (1987) from an Internet Archive disk-image item (single 720K disk, DOS)
S3	lesson screen with a practice line, an on-screen keyboard with drawn hands, gauges, a clock, a metronome, and a help box	same color scheme and origin as S2; attributed to Typing Tutor IV v1.0 (<i>Inference: the owner identified the multicolor screens as this program; S3 has no title of its own</i>)
S4	a falling-words game over a city skyline; typed prefix shown in red	program not identified
S5	a green-on-black falling-letters game with counters and a key-order strip	Typing Tutor III (1984), per the owner; the green monochrome display fits the catalog’s monochrome screen-shot set

18.3.1 S1 and S2: the packaging

The two title screens show that a falling-letter game was distributed *with* a lesson-based tutor, not separately, and that the pairing survived a revision: the menu of S1 offers to run the tutor or to play the game, and S2 keeps the pairing under a new number. In the taxonomy of Chapter 16 the bundle is a lesson-based (copying and generative) tutor paired with a gamified one.

18.3.2 S3: the lesson screen

The lesson screen of S3 is unusually legible. It shows a practice line (a trivia-flavored sentence) printed twice, one copy above the other, with a cursor after the lower copy, which suggests a model line and the learner’s echo of it. (*Inference: model and echo*) Beside the text are three vertical gauges with a numeral above them, an analog clock, and a metronome drawn as a pendulum. Below is a keyboard diagram with hands, and a help box explains that the clock is useful for timed exercises and that the metronome is on the right. The screen therefore presents together the model (copying), a running record (gauges), a clock (timed writing), and

a pacing signal (metronome). It is the workbook's cassette tape, timed writings, and printed drill lines collapsed onto one screen. The gauge labels are too small to read reliably and are not transcribed. **[Source needed: manual text naming the gauges]**

18.3.3 Catalog evidence for Typing Tutor III

An online software catalog (PixelatedArcade) lists *Typing Tutor III* for IBM PC compatibles, released in 1984 in the United States, developed by Kriya Systems, Inc. and published by Simon & Schuster, Inc., with screenshot sets for a CGA color display and for a monochrome display. This book uses only the catalog's captions, which describe the following screens: title; several introduction screens; setting up a new user, and selecting an existing user or creating one; a main menu; practice typing letters; a typing status report, a status report menu, and several graphs charting progress; and, for the game, an overview, letters to be typed before they reach the bottom, a letter being "zapped", words mixed in as the game progresses, and a summary of successes and failures. **[Source needed: view the screenshots themselves and the manual; the captions are a secondary description]**

Three consequences for the book, each resting on captions only. First, the tutor keeps *per-user records*, since users are created and selected, and it reports on them through status reports and graphs. It is a step toward the ledger of Chapter 27, though whether the underlying data are keystroke-level or session summaries is not known. Second, the game is not restricted to single letters: words are mixed in later, which means that the letters-to-words transition of S4 is a feature of at least this program's game as well. Third, the game ends with a summary of successes and failures, so its counters (compare S5) feed a report.

18.3.4 Descriptive record for Typing Tutor IV

Library and retail records describe Typing Tutor IV in terms that bear directly on the taxonomy of Chapter 16. They are secondary descriptions, drawn from catalog entries and a product listing, and are to be checked against the running program and the manual.

- On first use the program asks the learner to register a name and choose a proficiency level (beginner, two-finger typist, or touch typist), and up to eight people can register. **[Source needed: manual confirmation of the level names and the limit of eight]**
- The program tracks each person's progress and offers lessons and tests at an appropriate level. Tests gauge proficiency and identify weak points, and the program then offers lessons that concentrate on those weaknesses. If accurate, this places it among the *diagnostic* and *corrective* tutors of Chapter 16, and not among the fixed-lesson tutors.
- An IBM PC edition is described as adding an instructor's mode, more lessons, improved menus, and multiple speed tests, and its game as offering play at the learner's current level or with the full keyboard. **[Source needed: which edition and version this description belongs to]**
- Editions exist for several platforms, among them the IBM PC family, Apple IIgs, Macintosh, and Commodore 64 and 128.

The disk-image item from which S2 and S3 were obtained is a single 720K disk with the start file `tt.exe`, and its description notes that the program has trouble with recent dates and should be given a date from the 1980s or 1990s when it asks. That the program asks for a date at all is consistent with dated per-user records, which is the mechanism a progress report needs.

Because the program is reported to select lessons from a learner’s weak points, it is a test case for the definition of Chapter 17. If its selection depends on error and hesitation data and not merely on speed, it is fully adaptive in that sense; if it only raises the level after success, it is *V*-adaptive. A recorded emulated session, logged with the ledger schema of Chapter 27, can settle which.

18.4 Evidence Schedule

Primary evidence for each program should include screenshots or reconstructed layouts, manual text, the exercise corpus, scoring rules, and a recorded emulated session. Appendix A specifies the record for each.

Chapter 19

Lesson-Based DOS Tutors

19.1 Continuity with the Paper Tutor

A lesson-based tutor on a computer can be compared with the paper sources of Chapter 8 dimension by dimension. The features that the paper tutor supplied by people are now supplied by software.

Function	Paper tutor	Software tutor
Presenting the model	printed page	screen text
Checking correctness	learner, supervisor	comparison routine
Timing	stopwatch, tape	system clock
Advancing	supervisor, completion sheet	threshold on speed and accuracy
Record	typed page, stored	score file or nothing

19.2 What Automation Changed and What It Kept

Automation kept the graded key order, per-lesson vocabulary, and repetition of short units. It changed the observability of the learner: timing became available at the level of the keystroke, so the state S of Chapter 17 becomes measurable, though tutors differ in whether they use it. The paper tutor's page, which contained every uncorrected error, was typically replaced by a summary.

19.3 The Pacing Signal in Software

The workbook's pacing is a cassette running at a stated rate. In S3 the same function is a metronome drawn on the lesson screen, and the learner can see the beat while typing the model line. The transformation preserves the idea, an external signal for a rate the learner does not yet control, and changes two things: the rate can now be adjusted by the program, and the signal is visible instead of audible. Whether the metronome could be set to the stepped rates

of the workbook (25, 30, 35 and 40 words per minute) cannot be decided from the screenshot.
[Source needed: manual for **Typing Tutor IV**]

19.4 Threshold Rules

A common software advancement rule is a conjunction of thresholds,

$$J(S) = \mathbf{1}[\text{WPM}_{\text{net}} \geq v^* \wedge \text{errors} \leq e^*].$$

Such a rule is V -adaptive only in a weak sense: it gates but does not select. Chapter 26 contrasts it with selection by expected improvement.

Chapter 20

Letter Invaders

Falling-letter games convert latency into space. The vertical position of a character is the time remaining in which the learner may respond.

20.1 A Formal Model

Let a letter appear at the top of a screen of height H (in rows) and fall at a speed v rows per second. The learner must strike the correct key before the letter reaches the bottom. The available response time is

$$\tau = \frac{H}{v}.$$

A hit occurs if the response latency ℓ satisfies $\ell < \tau$ and the key is correct. If letters appear at rate ρ per second and v increases with level, then the difficulty is a function of (ρ, v) , and the tutor's whole pedagogy lies in how that function is chosen.

20.2 A Worked Example

Take a screen of $H = 24$ rows. At level 1 letters fall at $v = 6$ rows per second, so $\tau = 4$ seconds; at level 5 the speed is $v = 12$, so $\tau = 2$ seconds. A learner whose median latency on a home-row key is 0.6 seconds hits nearly every letter at both levels. A learner whose latency on the transition from an outer key to the opposite hand is 2.4 seconds will pass at level 1 and miss at level 5, though the same learner's average speed may be unremarkable. The game exposes exactly the tail of the latency distribution that a words-per-minute score averages away, which is the component H of the learner state.

20.3 The Speed Curve

The speed curve $v(n)$ over levels n determines the tutor's theory. A linear curve treats difficulty as uniform. A curve that accelerates late treats early levels as consolidation. A curve that depends on the learner's recent hit rate is V -adaptive in the sense of Chapter 17. [Source

needed: documented speed curve and level structure for a specific Letter Invaders program]

20.4 Two Documented Screens

S1 and S2 establish that *Letter Invaders* was the game bundled with Typing Tutor III (1984) and IV (1983 and 1987 on its title screen). S5, from Typing Tutor III, shows the parameters of the model above on a single screen.

- **Counters.** The header reads “Letters: 12” and “Hits: 8”, so the game reports letters released and letters struck, from which a hit rate is directly readable.
- **Free-falling letters.** Three letters (A, A, S) are shown at three heights in different columns, so the fall is spatial and the columns are arbitrary.
- **A floor.** A dithered bar at the bottom marks the boundary at which a letter counts as missed.
- **A key-order strip.** Along the top runs the sequence `ASDFJKL;HEITC.ORNZUWB,YXVPQMG?/4738295610`, with the first four keys, `ASDF`, highlighted. The strip is a curriculum: it lists the order in which the game admits keys, and the highlight marks the current set. (*Inference: that the highlight marks the admitted set, and that the step is four keys*)

20.4.1 The key order as a curriculum

The strip is a key-introduction order $K_0 \subset K_1 \subset \dots$ in exactly the sense of Chapter 8, with a striking difference. The home row (`ASDF`, then `JKL;`) is followed by `H`, `E`, `I` and `T`, then `C`, the period, `O` and `R`, then `N`, `Z`, `U` and `W`, and so on, with the digits last in a scrambled order. Geometry governs the first eight keys. Frequency appears to govern what follows, since `E`, `T`, `I`, `O`, `R` and `N` arrive within the first twenty, but not perfectly: `Z` arrives before `B`, `Y` and `G`. Measured against the word list of Chapter 22, taking prefixes of the strip four keys at a time, the effect is large.

Keys admitted	Last four added	Letters	$ V_i $	cov. %
4	A S D F	4	11	0.6
8	J K L ;	7	21	1.1
12	H E I T	11	373	20.5
16	C . O R	14	1 380	38.6
20	N Z U W	18	3 309	62.6
24	B , Y X	21	4 494	72.6
28	V P Q M	25	7 922	92.6

With eleven letters admitted, the strip’s order captures about 20.5% of rank-weighted frequency, against 3.3% for the geometry-first design order of Chapter 23 at the same eleven letters (Exercise 3). The letters chosen matter more than the position of the keys, at least for the size of the available vocabulary. The comparison is cautious: the strip is a single

unattributed screenshot, and the design order was not built to maximize vocabulary. It nevertheless shows that a tutor can choose between a geometric order and a frequency order, and that the choice changes what lessons can say. **[Source needed: confirm in the running program that the strip is the level order and that the step is four keys]**

20.5 Errors as Visible Threat

A delayed or missed keystroke has a spatial consequence: the letter reaches the bottom. Error is thereby made a visible threat and not a count. Wrong keys may do nothing, may cost points, or may reset progress, and each choice encodes a view of what an error is. **[Source needed: scoring and penalty rules for specific programs]**

20.6 Key Selection

The set of letters that may appear is a key set K , exactly as in Chapter 8. A game that draws only from the home row is the paced, spatialized counterpart of the first lesson, and a game that draws from the whole alphabet corresponds to the coverage exercises. The random draw distribution is a further choice: uniform draws teach positions, while draws from an English digraph distribution teach transitions.

Chapter 21

Racing, Shooting, and Adventure Tutors

21.1 Words per Minute as Locomotion

In racing tutors the learner's speed becomes the speed of a vehicle. If the vehicle's position x advances by each correct character, $x(t) = \alpha c(t)$ where $c(t)$ is the number of correct characters and α a scale, then the slope of the trajectory is proportional to instantaneous speed, and the race is a plot of WPM against time in physical form.

21.2 Recognition as Targeting

Shooting tutors turn recognition and key selection into targeting: the letter to be typed is the target, and the key is the trigger. The interesting design question is what happens on a wrong key, since a wrong key can be a miss, a wasted shot, or an ignored input.

21.3 Progress as Geography

Adventure tutors distribute exercises across a map, so curricular advancement becomes geographic access. The lesson order is then a graph whose edges are gated by performance thresholds. This makes the curriculum visible as terrain, and it also permits non-linear paths, a feature that paper tutors did not have.

21.4 From Letters to Words

Screenshot S4 shows a variant in which whole words fall toward a city skyline, with two counters at the top. The words vary in length and difficulty (for example **Sharp**, **Zodiac**, **Yellows**, **abstractors**), several are capitalized, and the word being typed shows its correct prefix in red. The design raises three points. The unit of practice is the word, so the game

exercises sequences and not isolated keys. Feedback is per prefix, which makes the partially typed word a visible state. And the vocabulary is broad, so the game is a coverage exercise in the sense of Chapter 16 and not a restricted-alphabet one. The program is not identified. [Source needed: title, publisher, and platform for S4]

21.5 What Each Game Spatializes

In each case the game spatializes some component of the state S : falling games spatialize latency, racing games spatialize velocity, shooting games spatialize discrimination, and adventure games spatialize progress. The taxonomy of Chapter 16 places them together as gamified tutors, but the component of S that they expose distinguishes them. [Source needed: documented examples of each type]

Study Questions, Part VI

1. Choose a DOS-era tutor and fill the evidence record of Appendix A for it.
2. For a falling-letter game with $H = 20$ rows and $\tau = 2.5$ seconds, what is v ? What happens to τ if v doubles?
3. Which component of S does each of the four game types expose?
4. Write a threshold rule $J(S)$ that is not V -adaptive.

Part VII

The Exercises

Chapter 22

Constrained Literature

A line such as `a sad lad asks dad;` is language generated under an expanding admissibility boundary. This chapter treats such lines as a genre.

22.1 The Phenomenon

If K_i is the current key set and $\Sigma(w)$ the characters a word w requires, the vocabulary available to a lesson author is

$$V_i = \{ w \in V : \Sigma(w) \subseteq K_i \}.$$

The author does not choose freely from English but from V_i , and the peculiar texture of early exercises is the visible shape of that restriction. In the first exercises of the Academy pages the restriction produces `ass lad add alf` and `flak alas fall asks` in Exercise 1 and the phrase `a sad lad asks dad;` that closes Exercise 2. In Exercise 2 it produces `gaff gash lags hall` and the marginal words `gald` and `jaffa`.

22.2 Two Regimes

Chapter 8 distinguished two regimes. In the *admissibility regime* the constraint is what can be spelled, and the author strains for words. In the *salience regime* the constraint is what will exercise the new key,

$$V_i^{\text{new}} = \{ w \in V_i : \Sigma(w) \cap N_i \neq \emptyset \}.$$

Exercise 13 is entirely in the salience regime: `zip zoo zig zag, zinc: zulu zero jazz:, quiz doze size hazy:, zeal lazy fizz fuzz:.` The sentences on the pages that continue Exercises 6 and later show the same regime at the level of the sentence: `Ilse who is a redhead is a selfish girl who likes a wild life;` is a sentence chosen so that every content word contains W or I, and its meaning is incidental.

22.3 Kinship with Constrained Writing

The genre is a relative of the lipogram, a text written without one or more letters. Georges Perec’s novel *La Disparition* (1969) avoids the letter E throughout, and the lessons here are the pedagogical inverse: they permit a small set and exclude the rest. The comparison is instructive because in each case the constraint is announced by the format and its effects are read in the prose. In a typing lesson the constraint is announced by the diagram at the top of the page.

22.4 The Successive Alphabet

The following passages are reconstructed, not transcribed. They show how a text can be written under a sequence of enlarged key sets. The first line is observed; the rest are constructed.

QWERTY	
a sad lad asks dad;	(observed, keys: A S D F J K L ;)
a glad lad has a flask;	(constructed, adds G H)
a sad lad has used a glass flask;	(constructed, adds E U)
a lad has a rose; dad asks for a jug;	(constructed, adds R O)
today a lad dared to ask dad;	(constructed, adds T Y)

The prose begins almost as incantation, grows faintly surreal, and resolves into ordinary language only when the alphabet nearly completes.

22.5 Two Constraints Running in Opposite Directions

An acrostic mnemonic restricts the initial letters of the words in a phrase. A restricted-alphabet lesson restricts all the letters. The first constraint runs from a target set to a phrase; the second runs from a permitted set to a vocabulary. Chapter 24 shows self-devised mnemonics of the first kind for the Dvorak keyboard, and the contrast clarifies what each constraint is for: the acrostic fixes a position in memory, and the lesson fixes a movement in the hand.

22.6 Measuring a Restriction

The size $|V_i|$ is a measurable property of a lesson. Appendix D gives the procedure, and Table 22.1 reports the result for the design key order of Chapter 23, for QWERTY and for Dvorak by position matching. Letter matching has the QWERTY column by definition. Counts are taken from two lists. The first is a cleaned copy of the google-10000-english list (9 486 entries after removing short tokens, place names, and brands), and the column “cov.” is the share of rank-weighted frequency (weight $1/\text{rank}$) that the admissible words capture. The second, used only in the text below, is a 370 105-entry list of English words that includes many rare and inflected forms.

Four findings follow, each subject to the caveat that the frequency list is drawn from web text and the key order is a design.

Table 22.1: Admissible vocabulary by exercise under the design key order. Letters only; punctuation keys are omitted from the “new” columns because they form no words.

Ex.	QWERTY (and letter matching)				Dvorak by position			
	new	$ V_i $	$ V_i^{\text{new}} $	cov. %	new	$ V_i $	$ V_i^{\text{new}} $	cov. %
1	asdfjkl	21	21	1.1	aehnostu	185	185	20.7
2	gh	36	15	1.5	di	477	292	32.1
3	eu	175	139	3.3	g	670	193	33.4
4	ro	524	349	14.7	pr	1840	1170	42.2
5	ty	1102	578	35.8	fy	2302	462	54.1
6	wi	1819	717	46.9	c	3418	1116	60.4
7	nm	4089	2270	70.6	bm	4951	1533	72.7
8	vb	5360	1271	79.5	kx	5471	520	75.1
9	c	7366	2006	89.8	jw	6009	538	81.6
10	qp	9139	1773	98.9	l	8463	2454	94.7
11	x	9370	231	99.7	qv	9370	907	99.7
12	-	9370	0	99.7	z	9486	116	100.0
13	z	9486	116	100.0	-	9486	0	100.0

1. **The first exercise.** Only 21 entries of the cleaned frequency list are spellable from the QWERTY home-row letters, and some of them are fragments (for example `ds1`); the same list yields 185 words for the Dvorak home-row set, including `the`, `to`, `on`, `that` and `not`. The 370 105-entry list gives 173 against 1 599, a ratio of about nine in both lists.
2. **Coverage.** The admissible words capture about 1.1% of rank-weighted frequency at QWERTY Exercise 1 and about 20.7% at the Dvorak counterpart. QWERTY does not reach a third of the weighted frequency until Exercise 5, while Dvorak by position exceeds it by Exercise 3.
3. **The advantage is early.** From Exercise 7 onward the two vocabularies are close, and at Exercises 8 to 10 QWERTY under the design order is ahead in coverage, because the design introduces N, M, V, B and C while position matching sends Dvorak to its bottom-row letters K, X, J and W. The effect belongs to the first six exercises.
4. **Punctuation positions.** Exercise 12 introduces no letter on QWERTY and Exercise 13 introduces none on Dvorak. In each case V_i^{new} is empty, so the salience regime cannot operate through words.

The expectation that the first-exercise vocabulary would be far larger for Dvorak is therefore supported for the first six exercises and reversed later in this particular order.

Chapter 23

The QWERTY Curriculum

This chapter states a QWERTY curriculum as a template instantiation. It is a design, not a report of the historical order, except where the two observed exercises fix an anchor. (*Design, not history: the order below is proposed; only Exercises 1, 2, 6 and 13 are anchored to the Academy pages*)

23.1 The Ladder Template

For an exercise i with new keys N_i and admitted set K_i , the template produces groups as follows.

- G1.** A chain of the keys in N_i (Exercises 1 and 2), or, for later exercises, a sandwich drill in which each new key is placed between two presses of the home key of its finger column (ded, juj).
- G2.** The chain with one old key inserted (Exercises 1 and 2), or the sandwich drill with the new letter capitalized, so that shift coordination is practiced from the first reach.
- G3.** A scramble of the first group, or short words from V_i^{new} .
- G4.** Words from V_i^{new} .
- G5.** Longer words from V_i^{new} .
- G6.** A phrase.
- G7+.** Sentences, in which each content word contains a new key where possible.

Each group is printed as three lines, each a window of fixed width on the endless repetition of a unit. The template has three parameters: the key-introduction order, the lexicon, and the layout header. Appendix G contains the corpora generated from the template for both layouts.

23.2 The Key Order

The order below is designed so that the four anchor exercises are respected and every letter is introduced by Exercise 13.

Ex.	New keys	Status
1	A S D F ; L K J	anchored
2	G H	anchored
3	E U	design
4	R O	design
5	T Y	design
6	W I	anchored
7	N M	design
8	V B	design
9	C ,	design
10	Q P	design
11	X .	design
12	/ and shifted punctuation	design
13	Z	anchored

23.3 Home Row, Reaches, Alternation

The first two exercises complete the home row. Reaches follow as sandwich drills of the form home, reach, home, exactly as in both paper sources (*sws*, *kik*, *aqa*). Alternation exercises then use words in which consecutive letters fall on opposite hands, and same-hand exercises use words typed entirely by one hand. The workbook's later lessons treat alternate-hand and one-hand words as topics in their own right, which supports treating hand alternation as a curricular category and not merely as an incidental property of words.

23.4 Digraphs, Punctuation, Capitals, Numbers

After the letters, the curriculum introduces frequent digraphs as units, then punctuation as keystrokes within sentences, then capitals through case-alternating drills, then the number row, and finally longer passages. The workbook places numbers and symbols in separate drill blocks (Lessons 6 to 10 and 11 to 15) and does so after the alphabet is complete, a sequencing that the design here follows.

23.5 Longer Passages

The final stage moves to continuous prose and to timed passages. For the timed passages the design adopts the scoring conventions found in the workbook: five-minute timings repeated twice, cumulative word counts printed at the right margin, and a policy of not correcting errors. Appendix B states the scoring rules.

Chapter 24

The Dvorak Curriculum

The same ladder can be applied to Dvorak, and there are two ways to do so. This chapter derives both from the QWERTY exercises and shows what each costs.

24.1 Position Matching

Position matching transposes the QWERTY key order by physical location, using the mapping of Chapter 12. Exercise 1 becomes the keys A O E U and S N T H. The chain groups transpose cleanly:

G1	aoeu snth aoeu snth ...
G2	aoeua snths aoeua snths ...
G3	eauo tshn eauo tshn ...

(derived by mapping the QWERTY lines `asdf ;lkj`, `asdfa ;lkj`, `dafs k;jl`; not observed.) The word groups do not transpose. The QWERTY words `ass lad add alf` map to `aoo nae aee anu`, which are not words. A position-matched lesson can keep the geometry and must therefore replace the lexicon.

24.2 Dvorak-Native Vocabulary

Regenerating the words from V_1 for the Dvorak key set $\{a, o, e, u, h, t, n, s\}$ gives real and frequent words, among them `the`, `that`, `than`, `then`, `thus`, `east`, `aunt`, `haunt`, `those`, `these`, `south`, `house`, `hasten`, `tenants` and `sunset`. Groups 4 and 5 can be filled with these, and a phrase such as `the tan hen sat on the nest` is available for Group 6. The QWERTY first exercise has no word as common as `the`. Table 22.1 in Chapter 22 quantifies the difference: 185 admissible words against 21 in the cleaned frequency list, and about a fifth of rank-weighted frequency against about one percent. The generated corpus of Appendix G shows what the difference looks like on the page. Exercise 1 for Dvorak by position begins with the chains above, then the words `the that not has`, then `that these than state`, then `states estate status season`.

24.3 The Cost of Matching by Position

Position matching breaks down at later stages, because the mapping sends some QWERTY new keys to punctuation.

Ex.	Dvorak keys	Effect on the lesson
1	A O E U S N T H	rich vocabulary; no semicolon in the lesson
2	I D	completes the home row; sentences become possible
6	comma, C	the comma is a punctuation key, so words cannot exercise it; sentences must
13	semicolon	no letter at all; the lesson can only be sentences with clauses

At Exercise 6 the new key W maps to the comma, so a sentence such as **the students, the tenants, count the stones** is needed to exercise it. At Exercise 13 the Z position holds the semicolon, so the salience regime requires syntactic structure in place of words, since no word contains a semicolon. Matching by position exposes a fact that matching by letter conceals, namely that the difficulty of a key depends on its role in language as well as its location.

24.4 Matching by Letter

The alternative is to keep the QWERTY letter order and let the geometry differ. The vocabulary V_i is preserved, so the lexicon of each lesson is the same across layouts, but the diagram changes and the learner's movement at each stage differs. The two approaches answer different questions: position matching asks what the same movements permit, and letter matching asks what the same words require.

24.5 The Letter Region

A useful figure for each layout is the outline of the letter region by row.

Row	QWERTY letters	Dvorak letters
Top	Q to P (10 keys)	P to L (7 keys)
Home	A to L (9 keys)	A to S (10 keys)
Bottom	Z to M (7 keys)	Q to Z (9 keys)

Dvorak concentrates the letters on the home row and places punctuation where the QWERTY letters were, so the region that a QWERTY typist reaches most often for letters is a region of punctuation in Dvorak.

24.6 Transfer Learners and Region Mnemonics

A person retraining from QWERTY to Dvorak differs from the novice of the paper tutors, because habits already exist and must change. A set of handwritten retraining notes shows how such a learner can structure the task.

- **Start from invariants.** The notes begin with the two letters whose positions do not change, A and M, and drill them in combinations: `a m am ma amma mama`. The learner thereby begins with keys that carry no interference.
- **Name the displaced keys.** The notes record the interference directly, for example that the key in the QWERTY F position is U and the key in the J position is H, with the instruction to use the home row.
- **Drill by vowel and consonant blocks.** The blocks `aoeui` and `dhtns` are typed in isolation before words.
- **Use vowel-free words as consonant drills.** `ssh, shh, nth, brr, hmm`.
- **Use rare-key words.** `kook, yippee, yuppie, exequy, opaque, upkeep` exercise the keys on the outer and lower rows.

The notes also record acrostic mnemonics that fix the columns by language.

Region	Keys (top to bottom)	Mnemonic
Inner left column	Y, I, X	<i>Yes, its xmas</i>
Inner right column	F, D, B	<i>find daily bread</i>
Right edge	L, S, Z	<i>la zy</i> (marks corners L and Z; S implicit)
Left edge	P, A, Q	none recorded

Each phrase fixes a column by the initials of its words read in top-to-bottom order, so the mnemonic is a positional index and not a drill. The right-edge mnemonic marks endpoints only. The gap on the left edge is the natural place to add a further aid. These are self-devised aids of a kind that no paper tutor supplies: the paper tutors teach location by repetition and by the highlighted diagram, and this learner adds a fourth route through language.

Chapter 25

Two Pedagogical Dialects

Two layouts generate different early English even when both train toward the same competence.

25.1 Parallel Passages

The QWERTY lines below are transcribed from the Academy pages. The Dvorak lines are constructed for illustration, using only the keys that position matching admits at the same exercise number.

Exercise 1	
QWERTY	ass lad add alf
Dvorak	the tenants hasten out at sunset

Exercise 2	
QWERTY	a sad lad asks dad;
Dvorak	the students hate it and sit in the sun

Exercise 6	
QWERTY	Ilse who is a redhead is a selfish girl who likes a wild life;
Dvorak	the students, the tenants, count the stones

The first Dvorak line is a plausible English clause and the first QWERTY line is a list of near-words. By Exercise 6 the new keys of the Dvorak lesson are a comma and C, so the comma has forced a sentence structure onto the lesson, while the QWERTY sentence is selected only to contain W and I.

25.2 Convenient versus Representative Text

A pedagogically convenient sequence is not necessarily a linguistically representative one. A line built from the home row of QWERTY is easy to generate and lexically odd; a line built from the home row of Dvorak is closer to ordinary prose, because English concentrates its frequent letters there. Yet representativeness is not always a virtue in a lesson. A representative lesson

exercises the most frequent transitions and may leave rare keys undertrained until late, while a lesson that is salient for the new key deliberately over-samples it. The two dialects show the tension: the geometry of a keyboard sets which sequences are easy to write, and the author's choice of regime decides whether the lesson follows that ease or corrects for it.

25.3 What the Dialects Reveal

Comparing the dialects makes visible three things that a single layout hides. The first is that the strangeness of early exercises is a property of the pair (layout, lexicon) and not of typing instruction as such. The second is that the difficulty of a key depends on its linguistic role, since a punctuation key cannot be exercised by a word list. The third is that the ladder is independent of the layout, which is why it can serve as the common baseline of the comparison.

Study Questions, Part VII

1. Write a five-line lesson under the alphabet $\{a, o, e, u, h, t, n, s\}$, then a second under the QWERTY home row, and compare their readability.
2. Use the script of Appendix D with a word list of your choice and report $|V_1|$ for both layouts. State the list.
3. Why can the comma not be exercised by a word list? What kind of lesson exercises it?
4. Devise a mnemonic for the left edge (P, A, Q) of the Dvorak letter region, in the manner of the three recorded in Chapter 24.
5. Compare position matching and letter matching. Which question does each answer?

Part VIII

Design

Chapter 26

Specification for an Ideal Tutor

The specification that follows collects the findings of the book into requirements. It is a design and makes no claim that any historical tutor met it.

26.1 Requirements

1. **Historical exercise modes.** The tutor supports the restricted-alphabet ladder of Chapter 8, the full-alphabet lesson template of the workbook (speed drill, keyboard drill, alphabet drill, feature drill, paced practice, timed writing), and coverage exercises.
2. **Both layouts.** The tutor accepts a layout as a parameter and regenerates the curriculum from the template of Chapter 23, offering position matching and letter matching as options (Chapter 24).
3. **Two error rates.** The tutor reports the uncorrected and the corrected error rates separately (Chapter 17), and offers a no-correction mode that reproduces the workbook's policy.
4. **Transition-specific hesitation.** The tutor maintains the matrix T_{ab} and selects transition drills for cells whose normalized latency is high.
5. **Rhythm without forced uniformity.** A pacing signal supplies a target band for inter-key intervals, not a single tempo, so that natural variation in the intervals of hard transitions is not penalized.
6. **A configurable falling-letter game.** A Letter Invaders descendant exposes its key set K , draw distribution, speed curve $v(n)$, and penalty policy as parameters (Chapter 20).
7. **Transfer support.** For a retraining learner the tutor begins with keys whose positions are unchanged, shows displaced keys explicitly, and accepts learner-authored mnemonics for regions of the keyboard.
8. **A complete ledger.** Everything above is recorded in the practice ledger of Chapter 27.

26.2 Selection

Lesson selection follows the optimization of Chapter 17. A tutor conforming to this specification is fully adaptive in the sense of that chapter: its selection depends on H , E , T and R and not only on V .

26.3 Threshold Rules Reconsidered

The workbook advances a learner by a fixed sequence with tests every fifth lesson. A software tutor can retain the tests but should not treat a threshold as the only signal. A test result is one entry in the ledger, and the learner's history of transitions and corrections is available to whoever, human or program, decides on advancement.

Chapter 27

The Practice Ledger

The paper tutor's typed page contains every uncorrected error, and most software tutors replace it with a number. Some kept more: the catalog description of Typing Tutor III (Chapter 18) mentions per-user records, status reports, and progress graphs, so the idea of a durable record was present in the DOS era in summary form. The ledger restores the page and adds what the page lacked: time.

27.1 Requirements

- **Append-only.** Entries are added and never altered.
- **Complete.** Every keystroke is recorded with the prompt position, the key struck, and a timestamp, including corrections.
- **Derivable.** Every summary, such as words per minute or an error rate, can be recomputed from the ledger, so that improvement can be inspected and not merely announced by a badge.
- **Renderable as a page.** The final text of any session can be printed as a typed page, so that a supervisor or the learner can read it as the paper tutor was read.

27.2 A Schema

A session is a header followed by a sequence of events.

```
{"session":"S0412","layout":"dvorak","lesson":"ex1","order":"position",
  "mode":"no-correction","pace":null}
{"t":0.000,"prompt_pos":0,"expected":"a","struck":"a"}
{"t":0.412,"prompt_pos":1,"expected":"o","struck":"o"}
{"t":0.981,"prompt_pos":2,"expected":"e","struck":"u"}
{"t":1.377,"prompt_pos":3,"expected":"u","struck":"u"}
```

From this record the tutor derives V (from counts and time), H (from the differences of t), E (from the pairs of expected and struck keys), T (from the intervals grouped by consecutive expected keys), and R (from correction events).

27.3 Ledger and Page

The ledger's page rendering is the direct descendant of the typed sheet filed behind a completion sheet. Where the sheet held every uncorrected error and no timing, the ledger holds both, so nothing that the paper tutor preserved is lost and the time dimension is added. The chapter therefore closes the argument of the book: the paper tutor was a program whose state was a stack of pages, and the ideal software tutor is one whose state is a complete, inspectable record.

Study Questions, Part VIII

1. Specify a schema field that would let the ledger distinguish a corrected error from an uncorrected one.
2. From the ledger schema, derive the matrix T_{ab} for one session.
3. Render a session as a page. What information does the page lose relative to the ledger?

Appendix A

DOS Tutor Evidence Record

For each DOS-era program documented in Part VI, record the following.

- Title, publisher, edition, and platform.
- Screenshots or reconstructed layouts of the menu, a lesson screen, a game screen, and a results screen.
- Manual text describing lessons, thresholds, and scoring.
- The complete exercise corpus for at least the first ten lessons, transcribed.
- Scoring rules and any advancement thresholds.
- A recorded emulated session with the ledger schema of Chapter 27 applied to it.
- The source of each item and the date it was obtained.

Records so far

- **Typing Tutor III with Letter Invaders** (Kriya Systems, 1984; Simon & Schuster). Have: title and menu screen (S1), and catalog captions listing the screens of the IBM PC version (users, practice, status reports, graphs, game overview and summary). Need: the screenshots themselves, the manual, exercise corpus, scoring rules, and the level structure behind the S5 key-order strip.
- **Typing Tutor IV with Letter Invaders** (Kriya Systems, 1983 and 1987 on the title screen; Simon & Schuster Software). Have: title screen (S2) and a lesson screen (S3), both from the v1.0 (1987) single-disk image described on the Internet Archive; library and retail descriptions of its adaptive behavior. Need: an emulated session recorded with the ledger schema, the manual, gauge labels, metronome settings, the Letter Invaders screen, and the lesson-selection rule.
- **Typing Tutor III, further screens:** S5, the letters game with its key-order strip, is from this program (1984). Need: the level structure behind the strip.
- **Unidentified falling-words game** (S4). Need: identification.

Appendix B

Scoring Rules

B.1 Speed

Gross and net speeds are defined in Chapter 17 with the five-character word. Timed passages in the workbook print a cumulative word count at the right of each line, so the word count at the end of a timing is read from the last completed line plus the count for the partial line, and speed is that count divided by the five minutes of the timing.

B.2 Workbook Drill Scale

On drill lines the errors in each set of lines are marked and counted, and the count is read against a scale.

Errors per set of lines	Reading
0 to 3	excellent
4 to 7	satisfactory
8 or more	extra practice

B.3 Timed Writings

Timed writings are five minutes, practiced once, then timed and repeated twice, scored by the supervisor. The thresholds applied at Progress Tests are not known from the pages available. **[Source needed: Progress Test page and Lesson Completion Sheet]**

Appendix C

Exercise Transcriptions

Each group is printed as three identical lines of the given unit, cycled to the line width and ending mid-unit. Units are transcribed once; line endings are noted where they differ. Lines that could be read only as show-through are omitted.

C.1 Academy Pages, Exercise 1 (p. 11), keys A S D F ; L K J

G1	asdf ;lkj	(ends: asd)
G2	asdfa ;lkj;	(ends: asd)
G3	dafs k;jl dafs k;jk dafs k;jl dafs k;jl ...	(ends: daf)
G4	ass lad add alf	(last unit: ass lad add ass)
G5	flak alas fall asks	(ends: ask)

C.2 Academy Pages, Exercise 2 (p. 12), new keys G H

G1	asdfg ;lkjh	(ends: asd)
G2	asdfgfa ;lkjhj;	(last unit: ;lk asd)
G3	gaff gash lags hall	(ends: lag)
G4	glad asks hash gala gald asks hash gala	(ends: ask)
G5	salad glass shall flash jaffa slash	(ends: lad)
G6	a sad lad asks dad;	(ends: sad)

C.3 Academy Pages, Exercise 6 (pp. 20, 21), new keys W I

G1	asdsWs ;lkikl	(ends: asdsWs;)
G2	AsDsWs ;LkIkL	(ends: As)
G3	wed kid was lie	
G4	ail law his owe	
G5	were soil will hide	(ends: sod)

```
G6 heir wear like wore          (ends: her)
G7 wise lied wild wire          (ends: lie)
G8 A wood fire is good; oil is worse for a fire; wood is for fire
G9 We killed wild doe for a while for food; Will would ride well
G10 A fellow said he walked a high hill; he grew riled if we rode;
G11 Ilse who is a redhead is a selfish girl who likes a wild life;
```

C.4 Academy Pages, Exercise 13 (p. 34), new key Z

```
G1 asaza ;l;p;                  (ends: asa)
G2 aSaZa ;L;P;
G3 zip zoo zig zag
G4 zinc: zulu zero jazz:
G5 quiz doze size hazy:
G6 zeal lazy fizz fuzz:
```

C.5 Academy Pages, an Exercise Between 6 and 13 (p. 31), Groups 7 to 13

```
G7 prove quote point quart print query      (ends: quest)
G8 quota press equip paste
G9 Peter and Paul ran quickly to the printers for a paper:
G10 King Pepe snapped his fingers as the Pekinese pup passed by;
G11 This lively path is paved and has a border of crimson poppies;
G12 I will quit the room again if he does not answer all queries:
G13 The Queen was presented with two quills at the banquet today;
```

C.6 Workbook, Lesson 1 (p. 1-1)

```
Speed Drill (each line three times)
The machine is of the same high quality.
See that it is cut from the second list.
The amount of the bank balance is a surprise.
It will be well to place the paper on record.

Keyboard Drill (each line twice)
a;sldkfjgh ghfjdksla; a;sldkfjgh ghfjdksla; a;sldkfjgh
aqa sws ded frf ftf fgf ;p; lol kik juj jyj jhj aza sxs
dcd fvf fbf ;/; 1.1 k,k jmj jnj
aqa aqua sws sail ded dead frf rain ftf tall fgf gulf juj
undo jyj year jhj huge kik kilo lol lock ;p; paws aza zoom
sxs exited dcd cuts fvf vain fbf bore jnj noon jmj move
k,k asks 1.1 Rd. ;/; a/c
```

Alphabet Drill (each set of lines twice)

a b c d e f g h i j k l m n o p q r s t u v w x y z

ab cd ef gh ij kl mn op qr st uv wx yz

abc def ghi jkl mno pqr stu vwx yz

and but can did eye for get him ice jam kit led me nip oil

pad quit rob say ton use vat why axe yet zoo

Only a few quite exceptional people do realize that there is
very little value in just talking without being able to say
something worthwhile.

Word Families, Phrases, Common Word Sentence (each line once)

sell sells sold; like likes liked; make makes made; sew sews
sewed; turn turns turned; write writes written; act acts acted

of the/in the/to the/but the/on the/and the/of the/we are/he
is/they are/I was/you were/it will/it might be/you will find/

He is planning to travel to the west coast early in October.

John was able to ask Robert to list all the supplies I need.

I think those skates will be advertised in the paper Monday.

There are not that many large parks close to where he lives.

Appendix D

Notation and Procedures

D.1 Notation

Symbol	Meaning
K_i	set of keys admitted by exercise i
N_i	new keys, $K_i \setminus K_{i-1}$
$\Sigma(w)$	set of characters required to type word w
V_i	vocabulary available at exercise i
V_i^{new}	words in V_i using at least one new key
G_j, E_i	group j ; exercise i
$S = (A, V, H, E, T, R)$	learner state: accuracy, velocity, hesitation, error distribution, transition competence, correction behavior
τ	response time available in a falling-letter game
J	advancement judgment

D.2 Computing the Vocabulary of a Lesson

The following short script computes V_i and V_i^{new} from a plain word list with one lowercase word per line. The key sets are supplied as strings of characters.

```
import sys

def vocab(words, K):
    return [w for w in words if set(w) <= set(K)]

def vocab_new(words, K, N):
    return [w for w in vocab(words, K) if set(w) & set(N)]

# QWERTY position-to-Dvorak mapping, by physical key
Q = "qwertyuiopasdfghjkl;zxcvbnm,./"
D = "'.pyfgcrlaoeuidhtns;qjkbmwnvz"
to_dvorak = dict(zip(Q, D))
```

```
def transpose(text):
    return "".join(to_dvorak.get(c, c) for c in text)

if __name__ == "__main__":
    words = [w.strip() for w in open(sys.argv[1]) if w.strip().isalpha()]
    K1_qwerty = "asdfjkl;"
    K1_dvorak = transpose(K1_qwerty)
    print(len(vocab(words, K1_qwerty)), len(vocab(words, K1_dvorak)))
```

The script is a specification of the measurement in Chapter 22. The counts it returns depend on the word list, which must be stated with any reported figure.

Appendix E

Glossary

Admissible vocabulary. The words that can be typed using only the keys admitted so far.

Chord. A set of keys pressed together as one stroke.

Coverage exercise. A text constructed to contain every letter, such as a full-alphabet sentence.

Ladder. The sequence of groups in an exercise, from chains of the new keys to sentences.

Legibility. The property of a tutor whose pedagogical rule can be recovered from its interface.

Position matching. Transposing a lesson from one layout to another by physical key location.

Sandwich drill. A drill in which a reached key is placed between two presses of a home-row key.

Stroke. The unit of action in a chorded system.

Uncorrected error rate. The fraction of final characters that are wrong.

Corrected error rate. The fraction of keystrokes that were errors, whether or not repaired.

Appendix F

Evidence Register

Source	Status	Note
Academy pages 1, 11, 12, 20, 21, 31, 34	photographed, transcribed	exercise number of p. 31 not established; other exercises absent
Workbook instruction pages i, ii, iii	photographed	text on the reverse partly legible as show-through
Workbook table of contents, Lessons 1 to 20	photographed	complete
Workbook Lesson 1, p. 1-1	photographed, transcribed	later lessons absent
Screenshots S1, S2	title screens, identified	Typing Tutor III and IV with Letter Invaders
PixelatedArcade catalog, Typing Tutor III	captions only, read 20 September 2026	1984, IBM PC compatibles, Kriya Systems, Simon & Schuster
Screenshot S3	lesson screen	Typing Tutor IV v1.0, per the owner
Internet Archive item, Typing Tutor IV v1.0 (1987)	catalog description	single 720K disk, DOS, start file tt.exe; date-entry caveat
Library and retail descriptions of Typing Tutor IV	secondary	adaptive behavior, registration, editions; to be verified
Screenshot S5	game screen, Typing Tutor III	strip read from the image; meaning inferred
Screenshot S4	game screen	program not identified
Handwritten Dvorak re-training notes	photographed	used as evidence in Chapter 24; the meaning of several small annotations was confirmed by the owner
Handwritten design notes for Chapter 8 and the QWERTY and Dvorak curriculum sheet	photographed	design, not evidence of any historical tutor

Outstanding Items

- Whether the Academy pages and the workbook shared a physical binder, and how a learner moved between them.
- A Lesson Completion Sheet and a Progress Test page.
- The reverse sides of Academy pages 20 and 34, and pages for the missing exercises.
- Later workbook lessons (numbers, symbols, alternate-hand and one-hand words, spelling).
- A curated, hand-checked word list to replace the web-derived list used in Chapter 22.
- Identification of screenshot S4.
- A recorded emulated session of Typing Tutor IV v1.0 and of Typing Tutor III.
- DOS-era program manuals and further screens, per Appendix A.
- Sources for every item flagged in the text.

Appendix G

Generated Corpora

These corpora instantiate the ladder template of Chapter 23 for the design key order. They are generated by a script, not transcribed from any tutor. Words are drawn in frequency order from the google-10000-english list (no-swears version), after removing tokens shorter than three letters (with a short whitelist of common two-letter words) and a blacklist of place names, brands, and fragments. Groups G3 to G5 take words that use at least one new key, ordered by rank, in length bands of three to four, four to five, and six to nine letters. G6 is a string of frequent admissible words and is not a grammatical phrase. The lists are a starting point for editing, and the leftover oddities (for example the “words” **ds1** and **aka** that survive in the frequency list) are a reminder that a web-derived list has its own admissibility problems. Each exercise prints every group as three lines in the paper tutor; here each group is shown once.

Chains (G1, G2) are sandwich drills using the home key of the same finger column. In the Dvorak corpus the chains are the QWERTY chains transposed by position, and capitals are applied only where the transposed key is a letter.

G.1 QWERTY Design Corpus

```
Exercise 1  new keys: ;adfkls
```

```
G1  asdf ;lkj
G2  asdfa ;lkj;
G3  all add ask fall
G4  fall falls adds salad
G5  falls salad
G6  as all add ask ad
```

```
Exercise 2  new keys: gh
```

```
G1  asdfg ;lkjh
G2  asdfgfa ;lkjhj;
G3  has had half gas
G4  shall half flash glass
G5  shall flash glass flags
G6  has had half gas hall
```

Exercise 3 new keys: eu

G1 ded juj
G2 dEd jUj
G3 use see used she
G4 used full sale sales
G5 useful league glasses sealed
G6 us he use see used

Exercise 4 new keys: or

G1 frf lol
G2 fRf lOl
G3 for are free our
G4 free here also good
G5 should address offers seller
G6 of for or are free

Exercise 5 new keys: ty

G1 ftf jyj
G2 fTf jYj
G3 the that you your
G4 that your other they
G5 health through results states
G6 the to that you at

Exercise 6 new keys: iw

G1 sws kik
G2 sWs kIk
G3 this with was will
G4 this with will site
G5 software rights details without
G6 is this with it was

Exercise 7 new keys: mn

G1 jnj jmj
G2 jNj jMj
G3 and not from new
G4 from more home time
G5 online system message general
G6 and in on not from

Exercise 8 new keys: bv

G1 fvf fbf
G2 fVf fBf
G3 have but web view
G4 have about view been
G5 business number available review
G6 by be have but web

Exercise 9 new keys: ,c

G1 dcd k,k
G2 dCd k,k

```
G3 can back city such
G4 which click back music
G5 search contact services service
G6 can back city such each
```

Exercise 10 new keys: pq

```
G1 aqa ;p;
G2 aQa ;P;
G3 page help top post
G4 page help price post
G5 people products product policy
G6 page up help top post
```

Exercise 11 new keys: .x

```
G1 sxs l.l
G2 sXs l.l
G3 next text box tax
G4 next index text linux
G5 example exchange executive excellent
G6 next text box tax fax
```

Exercise 12 new keys: /

```
G1 ;/;
G2 ;/;
G3 (no listed word contains a new key)
G4 (no listed word contains a new key)
G5 (no listed word contains a new key)
G6 (none)
```

Exercise 13 new keys: z

```
G1 aza
G2 aZa
G3 size zip zone zoom
G4 size zone zoom zero
G5 magazine magazines arizona brazil
G6 size zip zone zoom zero
```

G.2 Dvorak Corpus by Position Matching

Exercise 1 new keys: aehnostu

```
G1 aoeu snth
G2 aoewa snths
G3 the that not has
G4 that these than state
G5 states estate status season
G6 the to on that not
```

Exercise 2 new keys: di

G1 aoeui snthd
G2 aoeuiua snthdhs
G3 and this site his
G4 this site date into
G5 united students issues student
G6 and in is this it

Exercise 3 new keys: .g

G1 e.e hgh
G2 e.e hGh
G3 get good high sign
G4 good high using sign
G5 design things against hosting
G6 get go good high sign

Exercise 4 new keys: pr

G1 upu nrn
G2 uPu nRn
G3 are page our here
G4 page other their there
G5 support rights through report
G6 or are page our up

Exercise 5 new keys: fy

G1 uyu hfh
G2 uYu hFh
G3 for you your free
G4 your free they first
G5 history property different features
G6 of for you your if

Exercise 6 new keys: ,c

G1 o,o tct
G2 o,o tCt
G3 can city such each
G4 price city such each
G5 search contact products product
G6 can city such each car

Exercise 7 new keys: bm

G1 hbh hmh
G2 hBh hMh
G3 from more home but
G4 from more home about
G5 business system number message
G6 by be from more home

Exercise 8 new keys: kx

G1 uku uxu
G2 uKu uXu
G3 back next make book
G4 back next make books

G5 market feedback making marketing
 G6 back next make book take

Exercise 9 new keys: jw

G1 eje twt
 G2 eJe tWt
 G3 with new was what
 G4 with what which news
 G5 software within subject between
 G6 with new was we what

Exercise 10 new keys: 'l

G1 a'a sls
 G2 a'a sLs
 G3 all will only also
 G4 will only also help
 G5 online people health should
 G6 all will only also help

Exercise 11 new keys: qv

G1 oqo nvn
 G2 oQo nVn
 G3 have view over very
 G4 have view over video
 G5 services service available review
 G6 have view over very even

Exercise 12 new keys: z

G1 szs
 G2 sZs
 G3 size zip zone zoom
 G4 size zone zoom zero
 G5 magazine magazines arizona brazil
 G6 size zip zone zoom zero

Exercise 13 new keys: ;

G1 a;a
 G2 a;a
 G3 (no listed word contains a new key)
 G4 (no listed word contains a new key)
 G5 (no listed word contains a new key)
 G6 (none)

G.3 Letter Matching

Under letter matching the word groups G3 to G6 of the QWERTY corpus are reused unchanged, since V_i is the same. Only the chains differ, because the sandwich drills must be rebuilt from the home keys of the Dvorak layout for the physical positions that the letters

occupy there.